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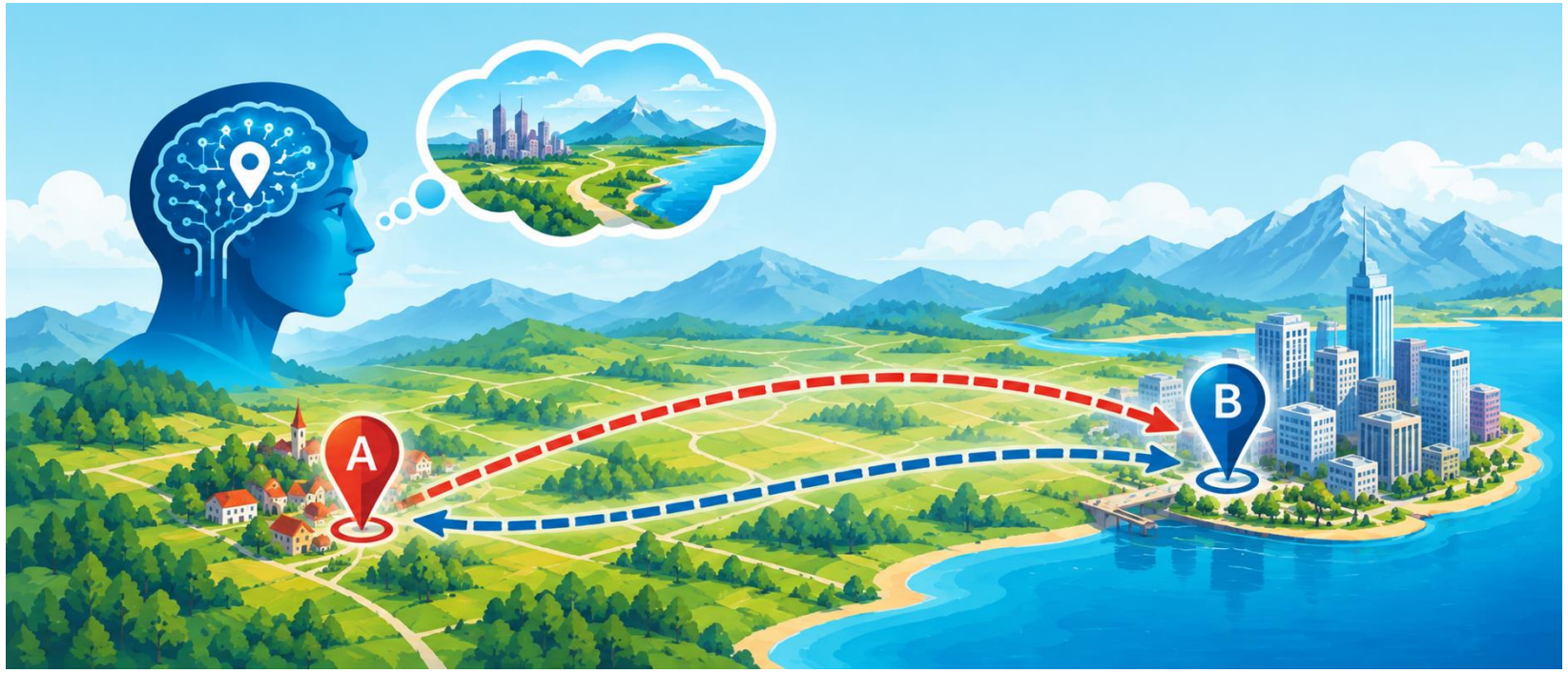
ISWC 2026 – SPARK 2026:
First International Workshop on Spatial Intelligence and
Reasoning Enabled by Knowledge Graphs and Foundation Models

ISWC BARI 2026

# Does Cognitive Plausibility Matter for Agentic GeoAI Persona?

## Testing Source-Target Asymmetry in Distance Estimation

26th October 2026 I Bari, Italy




Krzysztof Janowicz, **Alexandra Fortacz-Lazan,** Seyedeh Negin Kosari, Linus Lulkiewicz, Carl Feldmann, Sebastian-Paul Lambrecht, Bastian Maximilian Arnold, Fabio Spitzenstätter, Alexander Filipsky, Brenda Tarn, Nelly Dorrmann, Martin Jakob Winder, Paul Vitovec, and Sebastian Kerstein

Department of Geography and Regional Research
Spatial Data Science & Geocommunication

# Outline



Motivation



The Study



Experimental Set-up



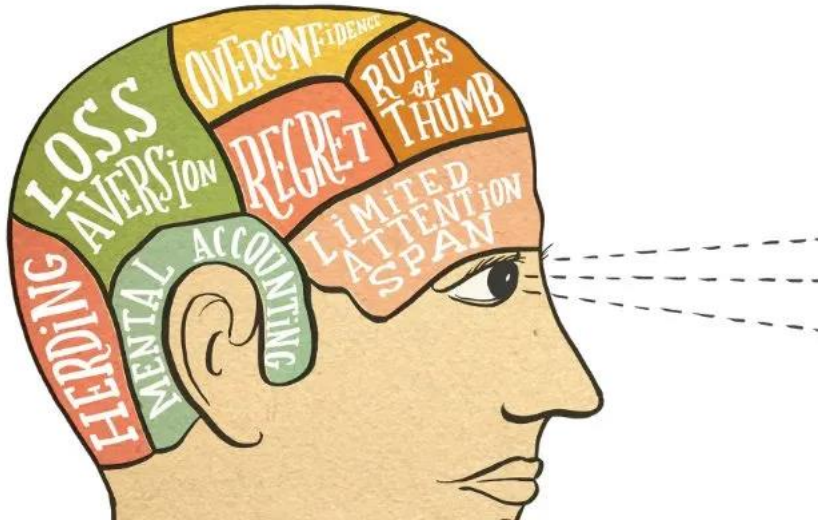
Results



Conclusions

# Motivations

- **AI agents** are *(Du et al., 2024)*
  - increasingly entering fields previously thought to require human-level intelligence
  - asked to simulated and approximate human behaviour, preferences, or even decisions
- Evaluation is focused on **accuracy**.
- **Unfortunately, this focus of accuracy may fall short for certain tasks:** e.g. human mobility, disaster mitigation (Hu et al., 2023), **…**
  - all involve perception, individual preferences, local knowledge, social context, and also **understanding cognitive heuristics**
  - Example: behavioural bias in economics *(Tversky & Kahnemann, 1974)*




Hunjunt , 2016: What is behavioral economics and why does it matter to marketers? 2026-09-16)

# Study I

- **Cognitive biases & heuristics in geography**: *asymmetry in distance estimations of landmarks* *(Sadalla, Burroughs, & Staplin, 1980)*

*"the cognitive distance between reference points and nonreference points is asymmetrical; nonreference points are judged nearer to reference points than are reference points to nonreference points."* *(Burroughs & Sadalla, 1979, p. 420)*

## Research Gap:

- Focus on accuracy of spatial abilities of AI
- Cognitive geographic bias and heuristics in AI?

***Is it sufficient to study only the ability of these systems and not also whether their behaviour is cognitively plausible?***

***Are AI agents able to mimic human behaviour*** *when required for specific tasks?*

# Study II

## Research Questions:

| | |
|---|---|
| ACCURACY | • Can **large language models (LLMs) accurately estimate the geographic distance** between source→target pairs latently, i.e., without lookup or using tools (e.g., reasoners)? |
| ASYMMETRY | • Do ordered source→target pairs remain invariant under reversal, i.e., **are distance judgments symmetric**? |
| SYSTEMATIC BIAS | • Is the lack of symmetry caused by prompt sensitivity and the probabilistic nature of generative AI, or **are outputs systematically biased in one direction**? |
| SUSTAINABILITY | • Can **smaller, and, hence, cheaper and potentially less resource-intensive models compete** with full SotA frontier models? |

# Experimental Set-up I

- Models:
  - GPT family: gpt-5.4-2026-03-05, gpt-4.1-2025-04-14, gpt-4.1-mini-2025-04-14, gpt-4.1-nano-2025-04-14, gpt-5.4-nano-2026-03-17,
  - a locally hosted gemma-4-31B-it developed by Google
- Test:
  - Source-target pairs: Top 500 cities from GeoNames by population *(major)* + 30 smaller cities (with a population over 1000) *(minor)* per top 500 city
  - 2 x 15,000 source and target city pairs
  - Temperature: 0.3
  - Reasoning effort for GPT-5 family: none
  - Token size limit: 20

# Experimental Set-up II

- Prompts:

Major-minor: *What is the distance between Shanghai, China and Würzburg, Germany?*

minor-Major: *What is the distance between Würzburg, Germany and Shanghai, China?*

**User prompt:** *What is the distance between {source city} and {target city}?*

**System prompt:** *Use your latent estimate. No lookup of the distances. Return only the distance in kilometers without units. Just return the plain number, e.g., 654.7, no text before or after.*

- Expectations:
  - Model distance estimates reflect the actual real distance estimates.
  - Distance estimates between major to minor cities are larger than the distance estimates between minor to major cities.
  - Smaller model results (e.g. GPT nano) can compete with larger model results.

# Results I: Accuracy (Error Estimation) & Sustainability

**Table 1**
Estimation error per model. Signed percentage error indicates systematic over/under-estimation, while absolute percentage error indicates the magnitude irrespective of direction.

| Model | Median/mean signed error (%) | Median/mean absolute error (%) |
|---|---|---|
| gpt-4.1 | -1.45 / -2.73 | 3.38 / 6.83 |
| gpt-5.4 | -1.06 / -2.08 | 3.44 / 7.59 |
| gemma-4-31B-it | 1.29 / 2.97 | 6.16 / 11.69 |
| gpt-4.1-mini | -2.23 / -1.81 | 7.04 / 11.94 |
| gpt-5.4-nano | -4.14 / -1.31 | 12.12 / 20.19 |
| gpt-4.1-nano | **3.37 / 27.00** | **21.02 / 46.89** |

- **Task performance:**
  - only 1% of answers were not usable;
  - big model performed well;
  - four of the six models tend to underestimate distances;
- ***Sustainability:*** smaller models perform substantially worse.
  - The open-weights model Gemma 4 performs well given that it can run on a single GPU workstation;

- **Human-like estimations:** In contrast to all other models, Gemma 4 did not use decimal places and only returned multiples of 5
- **Potential representation bias:** substantial differences in estimation quality (European to North American cities vs. Asian to African cities)

# Results II: Aysemmtry

**Table 2**
Relative directional asymmetry per model for source→target reversals.

| Model | Med./mean signed asymmetry (%) | Med./mean absolute asymmetry (%) |
|---|---|---|
| gpt-4.1 | 0.00 / -0.10 | 1.81 / 3.93 |
| gpt-5.4 | 0.12 / -0.31 | 2.06 / 4.84 |
| gemma-4-31B-it | **0.80 / 1.79** | 2.82 / 6.84 |
| gpt-4.1-mini | 0.20 / -0.48 | 4.69 / 9.02 |
| gpt-5.4-nano | 0.42 / 0.74 | 11.06 / 21.10 |
| gpt-4.1-nano | 0.16 / -0.87 | **12.13 / 42.68** |

→ *distance estimations are asymmetric.*

But, the signed results show that this effect is unlikely to reflect a strong systematic directional bias.

- **Possible alternative explanations for results:**
  - Prompt sensitivity
  - Temperature = unlikely

# Results +:

→ *Do estimates become more difficult as city distance increases?*

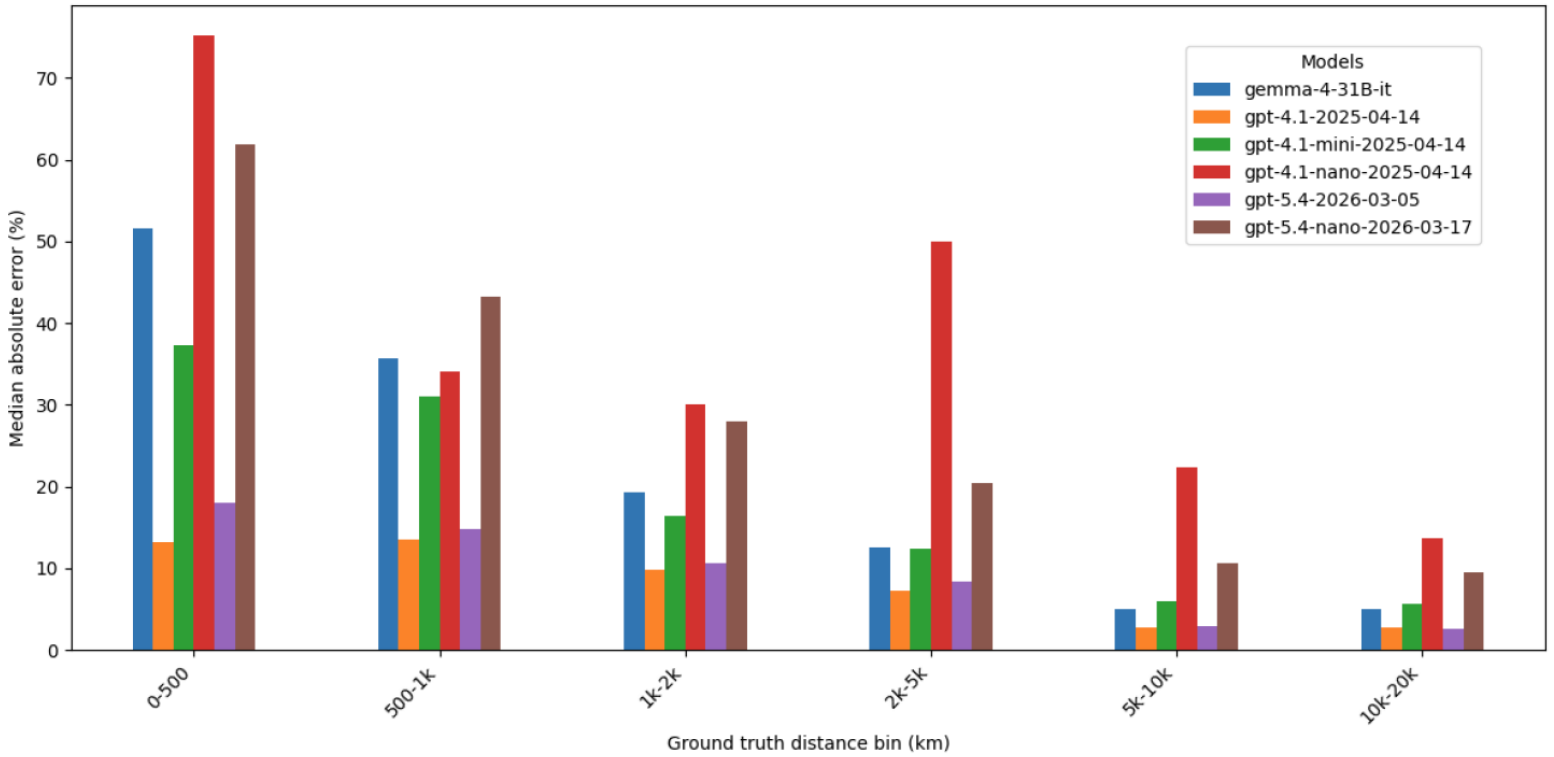



**Figure 1:** Distance estimation error by binned ground truth distance per model.

- **No -** as ground truth distance grows, the absolute percentage error shrinks (while the absolute error in km only increases modestly).
- Possible explanation: as models may lack detailed latent representation of most of the (very) small minor cities, models may use **country information**.

# Conclusions

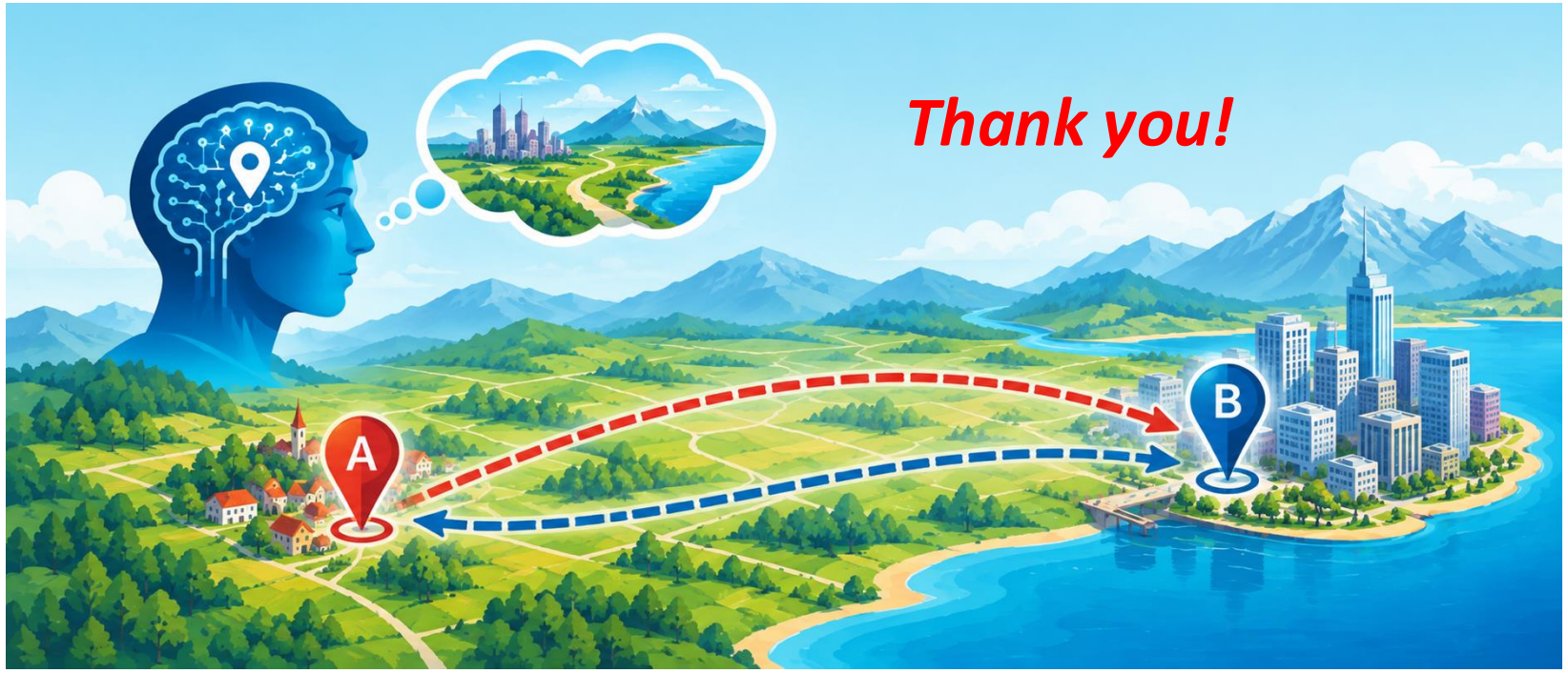
| | |
|---|---|
| ACCURACY | • LLMs can provide **overall accurate** geographic distance estimates |
| ASYMMETRY | • Yes, **results differ** under source→target pair reversal |
| SYSTEMATIC BIAS | • While distances are asymmetrical, they **do not follow the expected cognitive biases** and heuristics |
| SUSTAINABILITY | • **Differences** in accuracy between smaller and larger models are **very noticable** |

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# Result Evaluation

Mm = Major-minor direction
mM = Minor-major direction

$$\text{Signed percentage error} = 100 \cdot \frac{\hat{d}_i - d_i}{d_i}$$

$$\text{Absolute percentage error} = 100 \cdot \frac{\left|\hat{d}_i - d_i\right|}{d_i}$$

$$\text{Signed percentage asymmetry} = 100 \cdot \frac{\hat{d}_i^{Mm} - \hat{d}_i^{mM}}{d_i}$$

$$\text{Absolute percentage asymmetry} = 100 \cdot \frac{\left|\hat{d}_i^{Mm} - \hat{d}_i^{mM}\right|}{d_i}$$